IMPaCT-Data 2 - Deployment of the IMPaCT-Data Digital Platform to serve selected use cases

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IMPaCT Data

The second edition of the IMPaCT-Data program (IMPaCT-Data 2) focuses on assembling and deploying a **federated infrastructure** to standardize, integrate and analyze Clinical, Omics (primarily Genomics) and Medical Imaging Data.
This infrastructure will enable the use of biomedical data from ISCIII IMPaCT-Cohort and IMPaCT-PMP projects, using them to demonstrate the platform's utility, with the long-term aim to become a stable research infrastructure supporting biomedical research activities in Spain.

The **IMPaCT-Data 2** project is organized in 4 Work Packages:

- WP1 – Platform design & deployment
- WP2 – Use Cases & Interoperability
- WP3 – Innovative solutions
- WP4 – Coordination & Sustainability

AI + HPC + Benchmarking

Selected **USE CASES** are:

IMPaCT COHORTE + IMPaCT GENÓMICA

+

Precision Medicine Projects (PMPs):

INGENIO (Oncology)
Obes-IMPaCT (Metabolic)
VUScan (Hereditary cancer)
PREMYO (Cardiovascular)

Data (clinical, omic and image)
Research Requirements
Test final product usefulness

Users
- Healthcare Research Institutes
- Research Projects

Data
- Clinical
- Omics
- Image

Data Governance + ELSI

Platform

Frontend: **website**
- Tools + Workflows catalogues
- Service Portfolio

Backend: **key components**
- Federated Data Spaces
- Trusted Research Environment/s

Computing Centers

Authentication & Authorization

Reference Implementation – Key components

https://impact-data-ref-imp.readthedocs.io/en/

Data Storage	Local EGA Federated EGA
Discoverability	Beacon v2.0 Beacon Network Beacon OMOP Beacon cBioPortal
Data Analysis	Galaxy + Pulsar openVRE FEM + Flower
Security	Spanish National Security Schema (ENS) Keycloak (Authentication / Authorization)

For more details on the Reference Implementation Assessment, visit poster #69

All the developments, processes and requirements will be **aligned** and **interoperable** with relevant related initiatives

European Regulatory Framework and Associated Infrastructure

EHDS HealthData@EU Pilot

European Initiatives, Projects and Reference Infrastructures

European Genomic Data Infrastructure
EUCAIM CANCER IMAGE EUROPE
Federated European Genome-phenome Archive
B1MG
1+MG
B1MGplus
eosc
TEHDAS 2

National Projects and Initiatives

IMPaCT GENÓMICA
IMPaCT COHORTE
ISC Instituto de Salud Carlos III
INB Spanish National Bioinformatics Institute
ELIXIR SPAIN
TRANSBIONET Translational Bioinformatics Network

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European Genomic Data Infrastructure

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